SENIOR SCHOOL LEARNER'S NOTES

BASED ON THE KICD GRADE 10 CURRICULUM DESIGN

BIOLOGY

GRADE 10 ● SENIOR SCHOOL

A Learner's Textbook for Students Aged 14–16

Covering: Cell Biology & Biodiversity | Anatomy & Physiology of Plants | Anatomy & Physiology of Animals

Includes Idea Boxes, Case Studies, Reflection Questions, Quick Checks, Worksheets and a Complete Answer Key

TABLE OF CONTENTS

HOW TO USE THIS BOOK

This textbook is organised into three topics, matching the Grade 10 Biology curriculum. Each topic is divided into sub-topics containing the following features to support your learning:

- Learning Outcomes — what you should be able to do by the end of the sub-topic.
- Key Inquiry Question — a guiding question to think about as you study.
- Idea Box — an interesting fact related to the sub-topic.
- Case Study — a real-life example, often from Kenya, showing how the Biology you are learning applies to the real world.
- Reflection — an open question for you to think about or discuss; there is no single 'correct' answer.
- Quick Check — short questions to test your understanding immediately after each sub-topic.
- End of Topic Review Questions — a longer set of questions covering the whole topic.
- Practical Worksheet — a hands-on activity to help you apply what you have learnt.

All answers to the Quick Check and End of Topic Review Questions are provided in the Answer Key chapter at the very end of this book. Always attempt the questions on your own first before checking your answers — this is how you build strong understanding.

TOPIC 1

CELL BIOLOGY AND BIODIVERSITY

This topic introduces you to Biology as a science, how biologists collect and study living things, the cell as the basic unit of life, and the chemical substances that make life possible. By the end of this topic you will appreciate why Biology matters in everyday life and in various careers.

1.1 Introduction to Biology

Suggested Lessons: 6

Key Inquiry Question: *Why is it important to study Biology?*

By the end of this sub-topic, you should be able to:

- explain the application of Biology in everyday life
- relate fields of study in Biology to career opportunities
- illustrate the careers related to fields of study in Biology
- appreciate the importance of Biology in everyday life

What is Biology?

Biology is the branch of science that deals with the study of living organisms — their structure, function, growth, origin, evolution and distribution. The word 'Biology' comes from two Greek words: bios (life) and logos (study). Anything that is alive — from the smallest bacterium to the largest whale — is studied under Biology.

Biology touches almost every part of our daily lives. It helps us understand our own bodies, grow enough food to feed our families, treat and prevent diseases, conserve our environment, and develop new technologies such as vaccines and improved crop varieties.

Application of Biology in Everyday Life

- Agriculture — improving crop yields, animal breeding, pest and disease control
- Medicine and Health — diagnosis and treatment of diseases, drug development, vaccination
- Environmental conservation — protecting forests, wildlife and water catchments
- Food production and preservation — fermentation, food safety, nutrition
- Biotechnology — genetic engineering, production of insulin, tissue culture
- Forensic science — using DNA evidence to solve crimes

Fields of Study in Biology and Related Careers

Biology is a broad subject that branches into many specialized fields. Each field opens up specific career opportunities:

Field of Study	What it Studies	Related Career
Botany	Plants	Botanist
Zoology	Animals	Zoologist
Taxonomy	Classification of organisms	Taxonomist
Anatomy	Internal structure of organisms	Anatomist
Physiology	Functioning of body parts	Physiologist
Ecology	Organisms and their environment	Ecologist
Biochemistry	Chemical processes in living things	Biochemist
Biotechnology	Use of living things in technology	Biotechnologist
Genetics	Heredity and variation	Geneticist
Parasitology	Parasites and parasitic diseases	Parasitologist
Microbiology	Micro-organisms	Microbiologist
Entomology	Insects	Entomologist

Choosing a Career Wisely

When choosing a career related to Biology, you should be guided by your interest and ability. These are the factors that should influence your choice.

Some factors should NOT influence your career choice. These include gender, culture, disability, environment and stereotypes. For example, a girl can become a great zoologist and a boy can become a skilled nurse — Biology-related careers are open to everyone regardless of gender.

💡 Did You Know?

The Kenyan Nobel Laureate Prof. Wangari Maathai studied Biology (Anatomy) before founding the Green Belt Movement, which has planted over 50 million trees across Kenya. Her biological knowledge of plants and ecosystems helped her fight for environmental conservation and women's empowerment.

📖 Case Study: Prof. Wangari Maathai — From Biology Student to Nobel Laureate

Wangari Maathai studied Biological Sciences and later specialised in Veterinary Anatomy, becoming the first woman in East and Central Africa to earn a PhD. She used her scientific training to start the Green Belt Movement in 1977, mobilising communities — especially women — to plant trees to fight deforestation, soil erosion and poverty. In 2004, she became the first African woman to win the Nobel Peace Prize.

Discussion Questions:

1. Which field of Biology is most closely related to Prof. Maathai's environmental work?
2. Identify two skills a person needs to succeed in an ecology-related career.
3. In groups, discuss how studying Biology can help solve environmental problems in your own community.

🤔 REFLECTION — Think About It

Think about a career you would like to pursue in future. Which field of Biology is it related to, and why does it interest you?

✎ Quick Check 1.1

1. Define the term Biology.
2. State three applications of Biology in everyday life.
3. Match the following fields of study to their correct career: (a) Entomology (b) Microbiology (c) Genetics — Careers: Geneticist, Entomologist, Microbiologist.
4. Name two factors that should influence your career choice, and two that should not.

1.2 Specimen Collection and Preservation

Suggested Lessons: 14

Key Inquiry Question: *How are specimens collected and preserved?*

By the end of this sub-topic, you should be able to:

- identify apparatus and materials used for collecting, processing and preserving specimens
- collect, process and preserve specimens for biological studies using improvised and conventional apparatus
- appreciate the importance of collecting, processing and preserving specimens in Biology

Why Do Biologists Collect Specimens?

A specimen is a sample of an organism (or part of it) kept for study, identification, teaching or research. Biologists collect specimens to study their structure, identify unknown species, keep records of biodiversity, and use them as reference material or teaching aids.

Apparatus Used for Collecting Specimens

- Pooter/aspirator — for collecting small insects
- Pitfall trap — for trapping crawling insects
- Sweep net/aerial net — for catching flying insects
- Light trap — attracts and traps night-flying insects
- Tullgren funnel — extracts small organisms from soil/leaf litter using heat and light
- Pair of forceps, hand lens, knife/secateurs, digger, hand gloves
- Collecting bags, envelopes, labels, pencils/permanent ink pens, tracing paper

Processing and Preserving Specimens

Plant specimens are commonly preserved by making a herbarium. This involves pressing the plant flat, drying it, mounting it on a sheet, and labelling it with its common/local name, date and place of collection.

Animal specimens are processed by sorting, then preserved either by mounting on soft boards (for insects — pinning) or by wet preservation using preservatives such as ethanol (for soft-bodied animals). Each specimen must be clearly labelled.

Safety and Responsibility

When collecting specimens, always observe safety precautions (wear gloves, be careful with sharp tools) and collect only what is necessary — over-collecting can harm biodiversity. Always seek permission before collecting from protected areas.

💡 **Did You Know?**

A well-prepared and stored herbarium specimen can remain useful for scientific study for over 100 years! Some herbarium specimens in major museums date back to the 1700s.

📖 **Case Study: The East African Herbarium**

Housed at the National Museums of Kenya in Nairobi, the East African Herbarium holds over one million preserved plant specimens collected from across the region since the early 1900s. Scientists use these specimens to identify plant species, track changes in vegetation over time, and support conservation decisions.

Discussion Questions:

1. Why is it important to record the date and locality when preserving a specimen?

2. Suggest one way your school could start a small herbarium or insect collection responsibly.

🤔 **REFLECTION — Think About It**

Why do you think it is important to record the collection date and locality on every specimen label?

✎ **Quick Check 1.2**

1. List four apparatus used to collect insect specimens.
2. Outline the steps followed in making a herbarium.
3. State two preservatives commonly used for animal specimens.
4. Give two safety precautions to observe when collecting specimens.

1.3 Cell Structure and Specialization

Suggested Lessons: 30

Key Inquiry Question: *Why do plant and animal cells differ, and how are cells specialized?*

By the end of this sub-topic, you should be able to:

- differentiate between light and electron microscope
- describe the structure and functions of plant and animal cells as observed in an electron microscope
- prepare temporary slides for observation and estimation of cell size using a light microscope
- relate the structures of specialized cells in plants and animals to their functions
- appreciate the cell as the basic unit of life

The Cell — Basic Unit of Life

All living organisms are made up of cells. A cell is the smallest structural and functional unit of life. Some organisms, like bacteria, are made of a single cell (unicellular), while others, like human beings, are made of trillions of cells (multicellular).

Light Microscope vs Electron Microscope

Feature	Light Microscope	Electron Microscope
Source	Light rays	Beam of electrons
Magnification	Up to about 1500x	Up to over 1,000,000x
Resolution	Lower	Much higher (shows fine detail)
Specimen	Can be living	Must be dead (vacuum conditions)
Cost/Portability	Cheaper, portable	Expensive, laboratory-based

Resolution is the ability of a microscope to distinguish between two points that are very close together as separate points. The higher the resolution, the clearer and more detailed the image.

Structure of Plant and Animal Cells

Under the electron microscope, both plant and animal cells show a similar basic plan: a cell membrane enclosing cytoplasm, a nucleus, and organelles such as mitochondria. However, there are important differences.

Feature	Plant Cell	Animal Cell
Cell wall	Present (made of cellulose)	Absent
Chloroplasts	Present (for photosynthesis)	Absent
Vacuole	One large, permanent vacuole	Small, temporary vacuoles
Shape	Fixed, regular shape	Irregular, can change shape
Nucleus	Present, often pushed to a side	Present, usually central

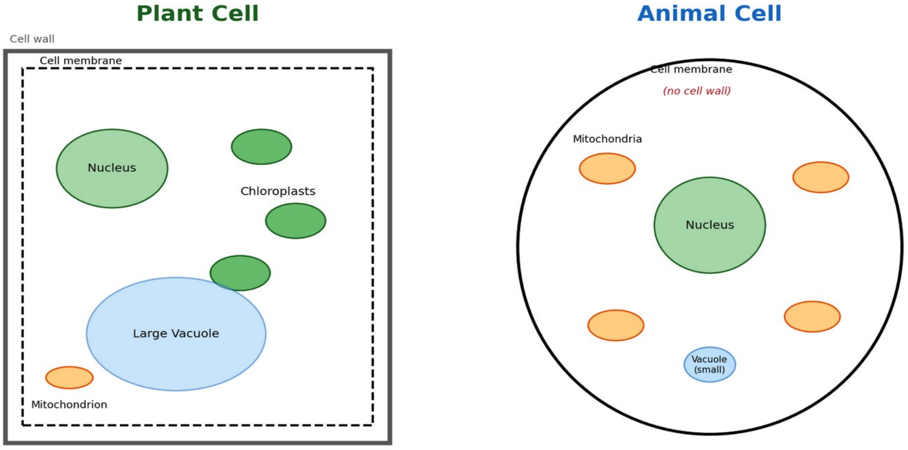

Fig 1.1: Structure of a plant cell and an animal cell

Preparing a Temporary Slide

To observe cells under a light microscope, you can prepare a temporary slide using a thin peel of onion epidermis or a young leaf. The general procedure is: obtain a thin section, place it on a slide with a drop of water or stain (e.g. iodine), cover gently with a coverslip avoiding air bubbles, and observe under low then high power. Cell size can be estimated by comparing it to the diameter of the microscope's field of view.

Specialized Cells

Cells become specialized (differentiated) — meaning they develop a particular shape and structure — to efficiently carry out a specific function.

Specialized Cell	Key Adaptation	Function
Root hair cell (plant)	Long thin extension, large surface area	Absorption of water and minerals
Palisade cell (plant)	Many chloroplasts, elongated shape near leaf surface	Photosynthesis
Guard cell (plant)	Kidney-shaped, can change turgidity	Controls opening/closing of stomata
Pollen grain (plant)	Tough outer wall, varied surface patterns	Carries male gametes for fertilization
Muscle cell (animal)	Long, contains contractile protein fibres	Contraction for movement
Nerve cell/neuron (animal)	Long axon, branched dendrites	Transmits electrical impulses
Red blood cell (animal)	Biconcave disc, no nucleus, contains haemoglobin	Transports oxygen
Sperm cell (animal)	Tail (flagellum), many mitochondria	Swims to and fertilizes the egg

Levels of Organization

In multicellular organisms, cells are organized into a hierarchy: Cell → Tissue → Organ → Organ System → Organism. A tissue is a group of similar cells performing the same function; an organ is made of different tissues working together; an organ system is a group of organs working together to perform a major body function.

> 💡 **Did You Know?**
>
> *The human body contains over 200 different types of specialized cells and an estimated 37 trillion cells in total! Yet every single one developed from just one fertilized egg cell.*

🤔 **REFLECTION — Think About It**

How does the long shape of a nerve cell (with a long axon) help it perform its function of transmitting messages?

✎ **Quick Check 1.3**

1. State two differences between a light microscope and an electron microscope.
2. List three structural differences between a plant cell and an animal cell.
3. Explain how a root hair cell is adapted to its function.
4. Arrange the following in the correct order of biological organization: Organ, Cell, Organ system, Tissue, Organism.

1.4 Chemicals of Life

Suggested Lessons: 28

Key Inquiry Question: *How are chemicals important in cells, and how is their presence determined?*

By the end of this sub-topic, you should be able to:

- describe the composition, properties and functions of the chemicals of life
- investigate the presence of carbohydrates, lipids, proteins and vitamin C in food substances
- investigate the presence of enzymes in living tissues
- determine factors affecting enzymatic reactions in cells
- appreciate the importance of chemical components in cells

The Chemicals of Life

Living organisms are made up of chemical substances that are essential for growth, energy, repair and normal functioning. The major chemicals of life are carbohydrates, proteins, lipids, vitamins, water, mineral salts and enzymes.

Chemical	Made of / Examples	Main Function
Carbohydrates	Sugars, starch, glycogen, cellulose	Main source of energy; cellulose gives plant cells structural support
Lipids (fats & oils)	Fatty acids and glycerol	Energy storage, insulation, forms cell membranes
Proteins	Chains of amino acids	Growth and repair; form enzymes, hormones and antibodies
Vitamins	E.g. Vitamin C, A, D	Required in small amounts for specific body processes; prevent deficiency diseases
Water	H2O	Solvent for reactions, transport medium, temperature regulation
Mineral salts	E.g. calcium, iron, potassium	Support processes like bone formation (calcium) and haemoglobin formation (iron)

Testing for Food Substances

Food Substance	Test	Positive Result
Reducing sugars	Benedict's test (add Benedict's solution, heat)	Colour changes from blue to green/yellow/orange/brick-red
Starch	Iodine test (add iodine solution)	Colour changes from brown to blue-black
Proteins	Biuret test (add Biuret solution/ NaOH + CuSO4)	Colour changes from blue to purple/violet
Lipids (fats/oils)	Emulsion test (shake with ethanol, then add water)	A cloudy white emulsion forms
Vitamin C	DCPIP test (add drops to DCPIP solution)	Blue DCPIP solution is decolourised

Enzymes

An enzyme is a biological catalyst — a protein that speeds up chemical reactions in cells without being used up. Enzymes are highly specific: each enzyme acts on a particular substrate.

One common enzyme is catalase, found in living tissues such as liver and potato, which breaks down hydrogen peroxide (a harmful by-product of metabolism) into water and oxygen. This can be demonstrated by adding fresh liver or potato pieces to hydrogen peroxide — vigorous bubbling (oxygen gas) shows the enzyme is active.

- Temperature — activity increases with temperature up to an optimum (usually around body temperature); beyond this, the enzyme is denatured and stops working
- pH — each enzyme has an optimum pH; activity falls sharply outside this range
- Substrate concentration — activity increases with more substrate, up to a point where all enzyme molecules are in use
- Enzyme concentration — more enzyme molecules generally increase the rate of reaction, provided there is enough substrate

Factors Affecting Enzyme Activity

Reading Food Packaging Labels

Packaging labels on food products show the quantity of nutrients, preservatives, colourings and the expiry date. Learning to read these labels helps you make informed and healthy choices when buying food.

> 💡 **Did You Know?**
>
> *Enzymes can speed up chemical reactions by up to a million times or more compared to the same reaction without an enzyme! Without enzymes, digestion of a single meal could take years.*

> 📖 **Case Study: Why Fever Can Be Dangerous**
>
> The human body normally functions at about 37°C, the optimum temperature for most human enzymes. During a very high fever (above 40°C), body enzymes — including those in the brain — can begin to denature, losing their shape and function permanently. This is why doctors treat high fevers urgently, especially in young children.
>
> **Discussion Questions:**
>
> 1. Explain, using the term 'denature', why a very high fever is dangerous.
> 2. Predict what would happen to enzyme activity if body temperature dropped far below normal (hypothermia).

> 🤔 **REFLECTION — Think About It**
>
> *Why do you think it is dangerous for a person to have a very high body temperature for a long time?*

> ✎ **Quick Check 1.4**
>
> 1. Name the food test used to detect the presence of starch, and state the positive result.
> 2. Define the term 'enzyme'.
> 3. State four factors that affect the rate of enzyme activity.
> 4. Explain why boiling a food sample before testing for catalase activity gives a negative result.

End of TOPIC 1 Review Questions

Answer all questions. Refer to the Answer Key chapter at the end of this book to check your work after attempting all questions on your own.

1. Define the term Biology and state two of its applications in everyday life.

2. Name any three fields of study in Biology and their related careers.

3. State two factors that should influence a person's choice of career, and two that should not.

4. List four pieces of apparatus used to collect biological specimens, stating what each is used for.

5. Describe the steps followed in preparing a herbarium specimen.

6. Distinguish between a light microscope and an electron microscope.

7. State three differences between a plant cell and an animal cell.

8. Relate the structure of a red blood cell to its function.

9. Arrange the following into the correct order of biological organization: tissue, organ system, cell, organism, organ.

10. Name the chemical test used to detect proteins and describe the expected positive result.

11. Define the term enzyme and state its main characteristic.

12. State four factors that affect the rate of enzyme-controlled reactions.

13. Explain why an enzyme stops working after it has been heated to a very high temperature.

14. State the functions of water and mineral salts in the body.

15. Explain the importance of reading food packaging labels before purchasing a food product.

Topic 1 Practical Worksheet: Testing Food Substances

Work in groups of four. Collect small samples of the following locally available foods: boiled potato, milk, cooking oil, mashed banana, and an orange. Carry out the Benedict's, iodine, Biuret and emulsion tests on each sample.

Tasks:

1. Prepare a results table with columns: Food Sample, Test Carried Out, Observation, Conclusion.

2. Record your observations for each test carefully, noting any colour changes.

3. State which food sample(s) tested positive for reducing sugars, starch, protein and lipids.

4. Suggest a reason why milk may test positive for more than one food substance.

5. Write a short report (half a page) summarising your findings and comparing them with another group's results.

TOPIC 2

ANATOMY AND PHYSIOLOGY OF PLANTS

In this topic you will learn how plants obtain food, how materials are transported within them, and how they exchange gases and respire. This knowledge is essential for understanding agriculture, food production and plant conservation.

2.1 Nutrition in Plants

Suggested Lessons: 12

Key Inquiry Question: *How do plants obtain food?*

By the end of this sub-topic, you should be able to:

- describe types of nutrition in plants
- relate the structure of the chloroplast to its function in plant cells
- illustrate the light and dark stages of photosynthesis in plants
- appreciate the significance of photosynthesis in nature

Types of Nutrition in Plants

Most plants make their own food through photosynthesis. This is called autotrophic nutrition. However, some plants cannot make all their own food and depend on other organisms — this is heterotrophic nutrition.

Mode of Nutrition	Description	Examples
Autotrophic	Manufactures its own food using light energy (photosynthesis)	Most green plants
Parasitic	Obtains food from a living host, often harming it	Dodder, mistletoe, Striga (witchweed)
Saprophytic	Obtains food from dead and decaying matter	Moulds, mushrooms
Symbiotic	Lives with another organism, both benefit	Rhizobium bacteria in legume root nodules; lichens
Insectivorous	Traps and digests insects to supplement nutrients (esp. nitrogen) in poor soils	Venus flytrap, pitcher plant, sundew

Structure of the Chloroplast

The chloroplast is the organelle where photosynthesis occurs. It has a double membrane, a fluid-filled stroma, and stacks of membranous discs called grana (singular: granum) made up of thylakoids, where chlorophyll traps light energy.

The Process of Photosynthesis

Photosynthesis is the process by which green plants manufacture food (glucose) using carbon dioxide and water, in the presence of light energy and chlorophyll, releasing oxygen as a by-product.

Photosynthesis occurs in two main stages: the light stage (light-dependent reactions), which takes place in the grana and traps light energy to split water and produce ATP and oxygen; and the dark stage (light-independent reactions), which takes place in the stroma and uses the products of the light stage to convert carbon dioxide into glucose.

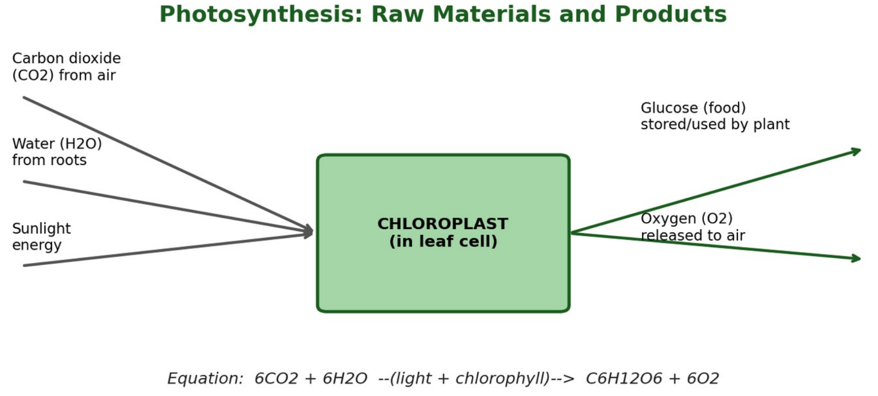

Fig 2.1: Raw materials and products of photosynthesis

Importance of Photosynthesis

- Produces food (glucose) for the plant and, indirectly, for all other organisms
- Releases oxygen used by living things for respiration
- Removes carbon dioxide from the atmosphere, helping regulate climate
- Forms the base of almost all food chains and food webs

💡 Did You Know?

Nearly all the oxygen you breathe was produced by photosynthesis — either by land plants or by microscopic marine algae called phytoplankton, which produce over half of the Earth's oxygen!

📖 Case Study: Insectivorous Plants in Nutrient-Poor Soils

Insectivorous plants such as the sundew and pitcher plant often grow in bogs and marshes where the soil is very poor in nitrogen. While these plants still photosynthesize to make sugars, they trap and digest insects to obtain extra nitrogen and minerals needed for making proteins.

Discussion Questions:

1. Explain why insectivorous plants are still considered mainly autotrophic.
2. Suggest why insectivorous plants are commonly found in nitrogen-poor soils rather than fertile farmland.

🤔 REFLECTION — Think About It

What do you think would happen to life on Earth if photosynthesis suddenly stopped everywhere?

✎ Quick Check 2.1

1. Distinguish between autotrophic and heterotrophic nutrition.
2. Give one example each of a parasitic and a symbiotic plant.
3. State the location and function of the grana in a chloroplast.
4. Write the general word equation for photosynthesis.

2.2 Transport in Plants

Suggested Lessons: 22

Key Inquiry Question: *How are materials transported in plants?*

By the end of this sub-topic, you should be able to:

- relate structures of the plant transport system to their functions
- illustrate the arrangement of vascular tissues in monocotyledonous and dicotyledonous plants
- demonstrate the uptake of water and mineral salts from the roots to the leaves
- demonstrate factors that affect the rate of transpiration
- describe the translocation of manufactured food in plants

The Plant Transport System

Plants have specialised tissues for transporting water, minerals and food. Roots absorb water and minerals through root hairs; the stem supports the plant and provides a pathway for transport; leaves are the main site of photosynthesis and gaseous exchange. The vascular tissues — xylem and phloem — run through roots, stems and leaves.

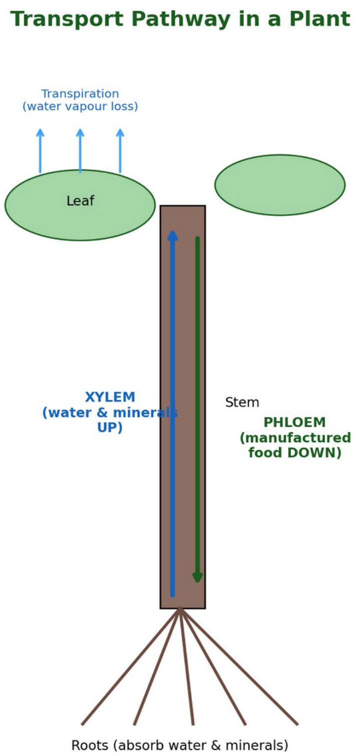

Fig 2.2: Pathway of water and food transport in a plant

Vascular Tissue Arrangement: Monocots vs Dicots

Feature	Monocotyledonous plants	Dicotyledonous plants
Vascular bundle arrangement (stem)	Scattered throughout the stem	Arranged in a ring near the edge
Example plants	Maize, grass, sugarcane	Bean, sunflower, mango

Uptake of Water and Mineral Salts

Water and minerals move from the soil into the root hairs and then upward through the xylem to the leaves. Three mechanisms assist this movement: root pressure (pushing water up from the roots), capillarity (the tendency of water to rise in narrow xylem tubes), and transpiration pull, which is the main driving force — as water evaporates from leaf surfaces, it pulls a continuous column of water up through the xylem (cohesion-tension theory).

Transpiration

Transpiration is the loss of water vapour from the aerial parts of a plant, mainly through the stomata. It helps cool the plant, maintains turgidity, and drives the transport of water and minerals.

- Light intensity — more light generally opens more stomata, increasing transpiration
- Temperature — higher temperature increases the rate of evaporation
- Humidity — lower humidity increases transpiration; high humidity reduces it
- Wind speed — moving air removes water vapour quickly, increasing transpiration
- Number and size of leaves — more leaf surface area increases the rate

Factors Affecting the Rate of Transpiration

Translocation

Translocation is the movement of manufactured food (mainly sucrose) from the leaves (the 'source') through the phloem to other parts of the plant where it is used or stored (the 'sink'), such as roots, fruits and growing shoots.

> 💡 **Did You Know?**
>
> *The tallest trees on Earth, like California's coast redwoods (over 100 metres tall), rely almost entirely on transpiration pull to move water from their roots all the way to their topmost leaves — against gravity!*

> 📖 **Case Study: Managing Water in Kenyan Agriculture**
>
> In arid and semi-arid areas of Kenya, farmers use knowledge of transpiration to manage crops efficiently — for example, by mulching (covering soil to reduce evaporation), planting windbreaks to reduce wind speed around crops, and irrigating during cooler parts of the day to reduce water loss.
>
> **Discussion Questions:**
>
> 1. Explain how mulching helps a crop conserve water.
> 2. Using your knowledge of factors affecting transpiration, suggest why farmers are advised to irrigate crops early in the morning or late in the evening.

> 🤔 **REFLECTION — Think About It**
>
> *Why do plants often wilt on a hot, dry and windy afternoon?*

> ✎ **Quick Check 2.2**
>
> 1. Distinguish between the arrangement of vascular bundles in monocots and dicots.
> 2. Name the three mechanisms that assist the upward movement of water in the xylem.
> 3. State four factors that affect the rate of transpiration.
> 4. Define translocation and name the tissue responsible for it.

2.3 Gaseous Exchange and Respiration in Plants

Suggested Lessons: 22

Key Inquiry Question: *Why is gaseous exchange important to plants, and how is respiration useful to them?*

By the end of this sub-topic, you should be able to:

- relate the structure of gaseous exchange sites in plants to their function
- describe the mechanism of opening and closing of stomata
- investigate aerobic and anaerobic respiration in living organisms
- explain the economic importance of anaerobic respiration

Sites of Gaseous Exchange in Plants

Site	Location	Notes
Stomata	Mainly on leaf surfaces (especially underside)	Main site of gas exchange; guarded by guard cells
Lenticels	On woody stems	Small pores for gas exchange in stems
Cuticle	Waxy layer on leaf surface	Very limited gas exchange, mainly prevents water loss
Pneumatophores	Aerial roots of mangroves	Allow gas exchange in waterlogged, oxygen-poor soils

Opening and Closing of Stomata

Stomata open and close due to changes in the turgidity of the guard cells surrounding them. When guard cells absorb water and become turgid, they bend outward, opening the stomatal pore. When they lose water and become flaccid, the pore closes. This movement is linked to the movement of potassium ions into and out of the guard cells, which affects water potential.

Aerobic and Anaerobic Respiration

Respiration is the process by which cells break down glucose to release energy. It can occur with oxygen (aerobic) or without oxygen (anaerobic).

Feature	Aerobic Respiration	Anaerobic Respiration
Oxygen required	Yes	No
Energy released	Large amount	Small amount
End products	Carbon dioxide + water	Ethanol + carbon dioxide (in plants/yeast)
Occurs in	Most living cells, most of the time	Yeast, some bacteria, muscle cells under low oxygen

Economic Importance of Anaerobic Respiration

- Baking — yeast ferments sugars, releasing carbon dioxide which makes bread rise
- Brewing — fermentation of sugars by yeast produces alcohol in beverages
- Biogas production — anaerobic bacteria break down organic waste to produce methane gas for cooking/lighting
- Silage making — fermentation preserves animal feed for use during dry seasons

- Production of yoghurt and lactic acid products through bacterial fermentation

💡 Did You Know?

Mangrove trees growing in waterlogged, oxygen-poor mud have special aerial roots called pneumatophores that stick up out of the mud like snorkels, allowing the roots to 'breathe' even when submerged!

📖 Case Study: Silage Making on Kenyan Dairy Farms

Many dairy farmers in Kenya's high-potential agricultural areas make silage by chopping fresh maize or grass and compacting it tightly in an airtight pit or bag. In the absence of oxygen, bacteria ferment the sugars in the plant material anaerobically, producing lactic acid which preserves the feed for months. This ensures a steady supply of nutritious feed for dairy cattle even during the dry season.

Discussion Questions:

1. Explain why the silage pit or bag must be airtight for the process to succeed.
2. Relate this practice to the economic benefits of understanding anaerobic respiration.

🤔 REFLECTION — Think About It

Why do bakers add yeast to bread dough before baking?

✎ Quick Check 2.3

1. Name four sites of gaseous exchange in plants.
2. Explain how guard cells control the opening and closing of stomata.
3. State two differences between aerobic and anaerobic respiration.
4. Give three economic uses of anaerobic respiration (fermentation).

End of TOPIC 2 Review Questions

Answer all questions. Refer to the Answer Key chapter at the end of this book to check your work after attempting all questions on your own.

1. Distinguish between autotrophic and heterotrophic nutrition, giving one example of each.

2. Describe the structure of a chloroplast and relate it to its function.

3. State the importance of photosynthesis to living organisms.

4. Distinguish between the arrangement of vascular bundles in monocotyledonous and dicotyledonous stems.

5. Explain three mechanisms that assist the upward movement of water in plants.

6. Define transpiration and state four factors that affect its rate.

7. Describe the process of translocation in plants.

8. Name four sites of gaseous exchange in plants and state where each is found.

9. Explain how the guard cells control the opening and closing of the stomata.

10. Distinguish between aerobic and anaerobic respiration.

11. Explain three economic importances of anaerobic respiration.

12. A farmer notices that his crops wilt faster on hot, windy afternoons than on cool, calm mornings. Explain this observation using your knowledge of transpiration.

Topic 2 Practical Worksheet: Investigating Transpiration

Working in groups, set up a potted plant experiment to investigate factors affecting transpiration using a plastic bag tied loosely over a leafy shoot, and repeat under different conditions (sunny/shaded, windy/still, warm/cool) using improvised materials such as a fan and a table lamp.

Tasks:

1. Record the time taken for visible water droplets to collect inside the polythene bag under each condition.

2. Present your results in a suitable table.

3. Identify which condition caused the fastest rate of transpiration and explain why.

4. Sketch and label the pathway of water from the soil to the atmosphere through the plant.

5. Suggest how farmers could apply your findings to reduce water loss in crops during dry seasons.

TOPIC 3

ANATOMY AND PHYSIOLOGY OF ANIMALS

This topic explores how animals feed, transport materials within their bodies, and exchange gases. You will discover how the structure of different body parts is closely related to the functions they perform — a key theme in Biology.

3.1 Nutrition in Animals

Suggested Lessons: 12

Key Inquiry Question: *How do insects and birds feed?*

By the end of this sub-topic, you should be able to:

- relate the structure of mouthparts of insects to their functions
- illustrate mouthparts in different insects
- relate the structure of beaks of birds to their functions
- appreciate diversity in feeding modes of insects and birds

Insect Mouthparts and Their Adaptations

Mouthpart Type	Example Insect	Adaptation & Function
Biting and chewing	Grasshopper, cockroach	Strong jaws (mandibles) for cutting and grinding solid plant material
Piercing and sucking	Mosquito	Needle-like proboscis pierces skin and sucks blood/fluids
Siphoning	Butterfly, moth	Long coiled tube (proboscis) sucks nectar from deep flowers
Cutting/sponging	Tsetse fly, housefly	Sharp mouthparts cut skin (tsetse fly) or sponge-like parts absorb liquid food (housefly)

Bird Beaks and Their Adaptations

Feeding Mode	Beak Adaptation	Example Bird
Grain/seed eating	Short, strong, conical beak for cracking seeds	Finch, weaver bird
Nectar feeding	Long, thin, curved beak to reach into flowers	Sunbird
Fish eating	Long, sharp, pointed beak for spearing/gripping fish	Kingfisher
Flesh eating	Strong, hooked beak for tearing flesh	Eagle, hawk
Filter feeding	Broad beak with comb-like filters	Flamingo
Wood chipping	Strong, chisel-like beak	Woodpecker
Multipurpose feeding	Medium, strong, general-purpose beak	Crow

💡 Did You Know?

Charles Darwin studied the beaks of finches on the Galapagos Islands and found that each species had a beak shape perfectly suited to the type of food available on its island — from cracking hard seeds to eating cactus flowers. This became one of the classic examples of adaptation in the history of Biology!

📖 Case Study: Darwin's Finches

On the Galapagos Islands, Charles Darwin observed several closely related species of finches, each with a differently shaped beak. Ground finches with large, strong beaks fed on hard seeds, while others with thin, pointed beaks fed on insects or cactus flowers. This variation showed how the structure of a beak matches its feeding function and habitat.

Discussion Questions:

1. Explain why a finch with a thin, pointed beak would struggle to crack open hard seeds.
2. Suggest what might happen to a bird population if the type of food available in its habitat suddenly changed.

🤔 REFLECTION — Think About It

How would a bird with a short, thick, seed-cracking beak struggle to survive in a habitat where the only food available is nectar from long tubular flowers?

✎ Quick Check 3.1

1. Name the mouthpart type found in a mosquito and state its function.
2. Match the following birds to their feeding mode: Kingfisher, Flamingo, Eagle, Sunbird.
3. Explain why a tsetse fly's mouthparts are adapted for cutting.
4. Give two examples of insects with biting and chewing mouthparts.

3.2 Transport in Animals

Suggested Lessons: 24

Key Inquiry Question: *Why is transport important in animals, and how do transport systems differ?*

By the end of this sub-topic, you should be able to:

- explain the importance of transport in animals
- illustrate structure of transport systems in insects, fish, amphibians, reptiles and mammals
- describe the pumping mechanism of the mammalian heart
- describe the human lymphatic and immune systems, and blood clotting mechanism
- explain the ABO and rhesus factor blood grouping systems in humans

Why Transport Is Important in Animals

- Carries digested food, oxygen and hormones to body cells
- Removes waste products such as carbon dioxide and urea from cells
- Helps distribute heat around the body
- Transports cells and substances involved in defence against disease

Transport Systems in Different Animal Groups

Animal Group	Type of System	Key Feature
Insects	Open circulatory system	Haemolymph (blood) flows freely in body cavity (haemocoel), not always confined to vessels
Fish	Closed, single circulation	Blood passes through the heart once per full circuit; two-chambered heart
Amphibians	Closed, double circulation (incomplete)	Three-chambered heart; some mixing of oxygenated and deoxygenated blood
Reptiles	Closed, double circulation	Mostly three-chambered heart with a partial septum (crocodiles have four chambers)
Mammals	Closed, double circulation (complete)	Four-chambered heart; oxygenated and deoxygenated blood kept fully separate

In an open circulatory system, blood is not always enclosed in vessels. In a closed circulatory system, blood remains within vessels (arteries, veins, capillaries) throughout the circuit — this is more efficient for larger, more active animals.

The Mammalian Heart and Its Pumping Mechanism

The mammalian heart has four chambers: two upper atria and two lower ventricles. Deoxygenated blood enters the right atrium from the vena cava, passes to the right ventricle, and is pumped to the lungs via the pulmonary artery. Oxygenated blood returns via the pulmonary vein into the left atrium, passes to the left ventricle, and is pumped out to the body through the aorta. Valves prevent backflow of blood, ensuring one-way flow.

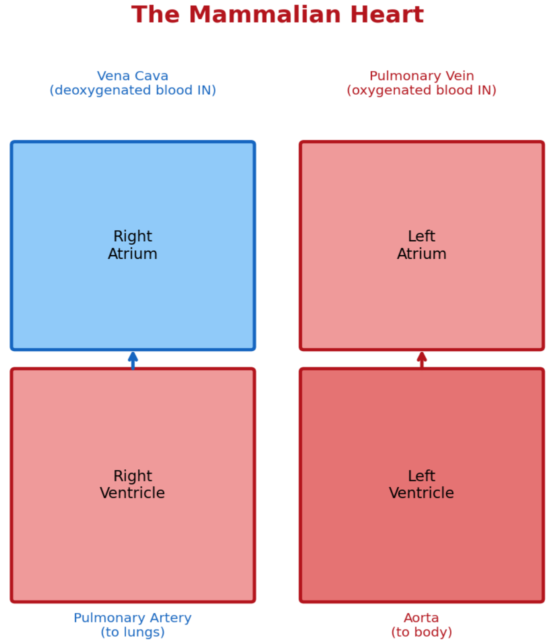

Fig 3.1: Structure of the mammalian heart

The Human Lymphatic and Immune Systems

The lymphatic system is a network of vessels and nodes that collects excess tissue fluid (as lymph) and returns it to the bloodstream. Lymph nodes contain lymphocytes (white blood cells) that help defend the body against infection — this links the lymphatic system closely to the immune system, which protects the body from disease-causing organisms.

Blood Clotting Mechanism

When a blood vessel is injured, platelets clump at the wound and release substances that trigger a series of reactions, converting the soluble protein fibrinogen into insoluble fibrin threads. These threads form a mesh that traps blood cells, forming a clot that seals the wound and prevents excessive blood loss and entry of pathogens.

ABO and Rhesus Blood Grouping

Human blood is grouped based on antigens present on red blood cells. The ABO system has four blood groups: A, B, AB and O. The Rhesus (Rh) factor further classifies blood as Rh-positive or Rh-negative. Compatibility between donor and recipient blood groups is critical during blood transfusion to avoid dangerous reactions.

Blood Group	Antigen on Red Cells	Can Donate To	Can Receive From
A	A	A, AB	A, O
B	B	B, AB	B, O
AB	A and B	AB only	A, B, AB, O (universal recipient)
O	None	A, B, AB, O (universal donor)	O only

💡 **Did You Know?**

People with blood group O-negative are called 'universal donors' because their blood can be safely given to patients of almost any blood group in an emergency, while AB-positive people are 'universal recipients' who can receive blood from any group.

📖 Case Study: Blood Donation Drives in Kenya

The Kenya Red Cross regularly organizes blood donation drives in schools, colleges and towns across the country to maintain a safe blood supply for hospitals. Donated blood is tested and grouped according to the ABO and Rhesus systems before being matched to patients in need — such as accident victims, mothers experiencing complications during childbirth, and patients undergoing surgery.

Discussion Questions:

1. Explain why donated blood must be tested and grouped before it can be given to a patient.
2. A patient with blood group A requires an emergency transfusion but blood group A is not available. Which other blood group could safely be given, and why?

🤔 REFLECTION — Think About It

Why is it important for every person to know their own blood group?

✎ Quick Check 3.2

1. Distinguish between an open and a closed circulatory system.
2. State the number of heart chambers found in a fish, an amphibian and a mammal.
3. Describe the pathway of blood through the mammalian heart, starting from the vena cava.
4. Explain the role of platelets in blood clotting.
5. State which blood group is known as the 'universal donor' and explain why.

3.3 Gaseous Exchange and Respiration in Animals

Suggested Lessons: 24

Key Inquiry Question: *How does gaseous exchange occur in animals, and why is respiration important?*

By the end of this sub-topic, you should be able to:

- explain the general characteristics of respiratory surfaces in animals
- describe the structure and adaptations of respiratory structures in animals
- describe the mechanism of gaseous exchange in humans
- describe the process of aerobic and anaerobic respiration
- calculate the respiratory quotient for different foods

Characteristics of Respiratory Surfaces

- Large surface area — allows more gas to diffuse at once
- Thin walls — short diffusion distance for gases
- Moist surface — gases dissolve before diffusing across
- Good blood supply — maintains a steep concentration gradient
- Permeable to gases

Respiratory Structures in Different Animals

Animal Group	Respiratory Structure	Notes
Insects	Tracheal system (spiracles, tracheae, tracheoles)	Air enters through spiracles and travels directly to tissues
Fish	Gills (gill filaments and lamellae)	Countercurrent flow of water and blood maximizes oxygen uptake
Amphibians	Lungs, skin and buccal cavity	Skin must stay moist for gas exchange to occur
Birds	Lungs with air sacs	Air sacs allow continuous, one-way airflow — very efficient
Mammals	Lungs (alveoli)	Millions of alveoli give a very large surface area for gas exchange

Gaseous Exchange in Humans

Air is drawn into the lungs during inhalation, when the diaphragm contracts and flattens and the rib muscles contract, increasing chest volume and drawing air in. During exhalation, the diaphragm relaxes and domes upward, and the chest volume decreases, pushing air out. Actual gas exchange occurs at the alveoli, tiny air sacs surrounded by capillaries, where oxygen diffuses into the blood and carbon dioxide diffuses out into the alveolus to be exhaled.

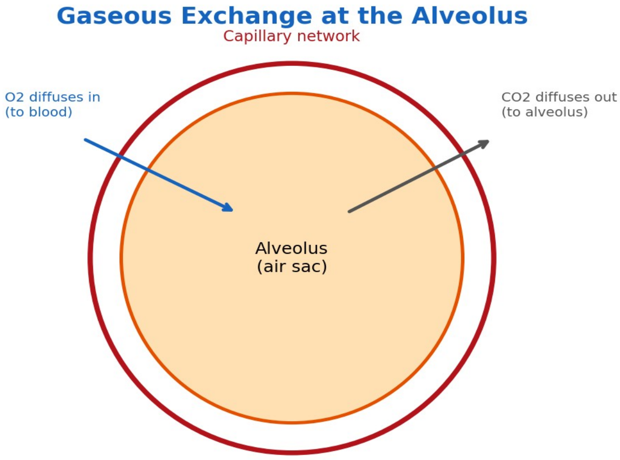

Fig 3.2: Gaseous exchange at the alveolus

Aerobic and Anaerobic Respiration in Animals; Oxygen Debt

Aerobic respiration uses oxygen to release a large amount of energy from glucose, producing carbon dioxide and water. Anaerobic respiration in animals occurs when oxygen supply is insufficient — for example, during vigorous exercise — and glucose is broken down to lactic acid, releasing a smaller amount of energy.

When muscles respire anaerobically for a period, lactic acid builds up, creating an 'oxygen debt'. This is why a person continues to breathe rapidly and heavily even after stopping exercise — the extra oxygen is needed to break down the accumulated lactic acid.

Factors Affecting Energy Requirements

- Age
- Sex
- Level of physical activity
- Body size
- Climate/environmental temperature

Respiratory Quotient (RQ)

The Respiratory Quotient is the ratio of the volume of carbon dioxide released to the volume of oxygen consumed during respiration: $RQ = CO_2 \text{ released} \div O_2 \text{ consumed}$.

Different respiratory substrates give different RQ values: carbohydrates give an RQ of about 1.0, fats about 0.7, and proteins about 0.8. The RQ value can therefore give a clue about which substrate is being respired.

> 💡 **Did You Know?**
>
> *Kenya's world-famous long-distance runners, many from the high-altitude regions around Iten and Eldoret, train where the air has less oxygen. Over time their bodies adapt by producing more red blood cells, improving their capacity to transport oxygen — giving them an advantage in oxygen-rich, lower-altitude race conditions.*

> 📖 **Case Study: High-Altitude Training in Kenya's Rift Valley**
>
> Towns like Iten, sitting at about 2,400 metres above sea level, have become world-renowned training centres for elite athletes. At high altitude, the air contains less oxygen, forcing the body to adapt by increasing red blood cell production and improving the efficiency of the respiratory and circulatory systems. When these athletes compete at lower altitudes, where oxygen is more plentiful, their enhanced oxygen-carrying capacity can give them a competitive advantage.
>
> **Discussion Questions:**
>
> 1. Explain why training at high altitude leads to an increase in red blood cell count.

2. Relate this case study to the concept of oxygen debt experienced during a sprint race.

🤔 REFLECTION — Think About It

Why does an athlete continue to breathe heavily for some time even after they have stopped running?

✎ Quick Check 3.3

1. State four characteristics of a good respiratory surface.
2. Name the respiratory structure used by fish and explain how it is adapted for gas exchange in water.
3. Explain the term 'oxygen debt'.
4. Calculate the RQ of a substrate that releases 20 units of CO_2 after consuming 20 units of O_2, and identify the likely substrate.
5. State two factors that affect energy requirements in humans.

End of TOPIC 3 Review Questions

Answer all questions. Refer to the Answer Key chapter at the end of this book to check your work after attempting all questions on your own.

1. Explain how the mouthparts of a mosquito are adapted to its feeding habit.

2. Relate the structure of an eagle's beak to its feeding mode.

3. State three functions of the transport system in animals.

4. Distinguish between an open and a closed circulatory system, giving one example of an animal with each.

5. Describe the pumping mechanism of the mammalian heart.

6. Explain the role of the lymphatic system in the human body.

7. Describe the mechanism of blood clotting in humans.

8. A road accident victim with blood group O is rushed to hospital and requires an urgent transfusion, but blood group O is temporarily out of stock. Explain whether blood group A, B or AB could safely be given.

9. State four characteristics of an efficient respiratory surface.

10. Describe how gaseous exchange occurs at the human alveolus.

11. Distinguish between aerobic and anaerobic respiration in animals.

12. Explain the term 'oxygen debt' and relate it to a real-life situation.

13. Define respiratory quotient and state its approximate value for carbohydrates, fats and proteins.

14. State four factors that affect energy requirements in the human body.

Topic 3 Practical Worksheet: Investigating Breathing Rate and Oxygen Debt

Work in pairs. Measure and record your partner's breathing rate (breaths per minute) at rest. Then have your partner perform 2 minutes of vigorous exercise (e.g. jogging on the spot or skipping), and immediately measure their breathing rate again, repeating the measurement every minute until it returns to the resting rate.

Tasks:

1. Present your results in a table showing time (minutes) against breathing rate.

2. Plot a simple line graph of breathing rate against time.

3. Describe the pattern shown by your graph.

4. Explain, using the term 'oxygen debt', why the breathing rate remained high for some time after exercise stopped.

5. Suggest one safety precaution that should be observed during this practical activity.

ANSWER KEY

This chapter provides model answers to all the Quick Check questions and End of Topic Review Questions in this book. Use it to check your own work — remember, some answers (especially longer explanations) can be correctly expressed in different words.

Quick Check Answers

Quick Check 1.1

1. Biology is the branch of science that deals with the study of living organisms.

2. Any three: agriculture, medicine/health, environmental conservation, food production, biotechnology, forensic science.

3. Entomology → Entomologist; Microbiology → Microbiologist; Genetics → Geneticist.

4. Should influence: interest and ability. Should NOT influence: gender, culture, disability, environment or stereotypes (any two).

Quick Check 1.2

1. Any four: pooter/aspirator, pitfall trap, sweep net, light trap, Tullgren funnel, forceps, hand lens.

2. Press the plant flat → dry it → mount it on a sheet → label with common/local name, date and locality → store and protect.

3. Ethanol (for wet preservation); formalin may also be accepted.

4. Any two: wear gloves, handle sharp tools carefully, avoid over-collecting, seek permission before collecting in protected areas.

Quick Check 1.3

1. Any two: light microscope uses light while electron microscope uses electrons; electron microscope has much higher magnification and resolution; electron microscope specimens must be dead while light microscope can view living specimens.

2. Any three: plant cell has a cell wall (animal doesn't); plant cell has chloroplasts (animal doesn't); plant cell has one large permanent vacuole (animal has small ones); plant cell has a fixed shape (animal cell shape can vary).

3. A root hair cell has a long, thin extension that increases its surface area, allowing more efficient absorption of water and mineral salts from the soil.

4. Cell → Tissue → Organ → Organ system → Organism.

Quick Check 1.4

1. Iodine test; the colour changes from brown to blue-black in the presence of starch.

2. An enzyme is a biological catalyst (a protein) that speeds up chemical reactions in cells without being used up.

3. Temperature, pH, substrate concentration, enzyme concentration.

4. Boiling denatures the enzyme (catalase), changing its shape so it can no longer act on its substrate (hydrogen peroxide); therefore no bubbling (oxygen release) is observed.

Quick Check 2.1

1. Autotrophic nutrition is when an organism manufactures its own food (e.g. by photosynthesis); heterotrophic nutrition is when an organism depends on other organisms for food.

2. Parasitic example: dodder, mistletoe or Striga. Symbiotic example: Rhizobium in legume root nodules, or lichens.

3. Grana are stacks of thylakoids inside the chloroplast; they contain chlorophyll and are the site of the light stage of photosynthesis.

4. Carbon dioxide + Water --(light energy + chlorophyll)--> Glucose + Oxygen.

Quick Check 2.2

1. In monocots, vascular bundles are scattered throughout the stem; in dicots, they are arranged in a ring near the edge of the stem.

2. Root pressure, capillarity and transpiration pull.

3. Any four: light intensity, temperature, humidity, wind speed, number/size of leaves.

4. Translocation is the movement of manufactured food (mainly sucrose) from the leaves to other parts of the plant; it occurs through the phloem.

Quick Check 2.3

1. Stomata, lenticels, cuticle, pneumatophores.

2. When guard cells take up water and become turgid, they bend outward and open the stomatal pore; when they lose water and become flaccid, the pore closes.

3. Any two: aerobic respiration requires oxygen while anaerobic does not; aerobic releases more energy than anaerobic; aerobic produces CO2 and water while anaerobic (in plants) produces ethanol and CO2.

4. Any three: baking, brewing, biogas production, silage making, yoghurt/lactic acid product manufacture.

Quick Check 3.1

1. Piercing and sucking mouthparts; used to pierce the skin and suck blood or plant fluids.

2. Kingfisher — fish eating; Flamingo — filter feeding; Eagle — flesh eating; Sunbird — nectar feeding.

3. The tsetse fly feeds on blood, so its mouthparts are sharp and adapted for cutting through the skin of its host to reach blood vessels.

4. Grasshopper and cockroach.

Quick Check 3.2

1. In an open circulatory system, blood (haemolymph) flows freely within the body cavity and is not always confined to vessels (e.g. insects); in a closed system, blood remains within vessels throughout the circuit (e.g. mammals).

2. Fish: two chambers. Amphibian: three chambers. Mammal: four chambers.

3. Vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary vein → left atrium → left ventricle → aorta → body.

4. Platelets clump at the site of injury and trigger reactions that convert fibrinogen to fibrin, forming a mesh that traps blood cells and seals the wound.

5. Blood group O (specifically O-negative) is the universal donor because its red blood cells have no A or B antigens, so it is unlikely to trigger an immune reaction when given to patients of other blood groups.

Quick Check 3.3

1. Large surface area, thin walls, moist surface, good blood supply (any four, permeability also acceptable).

2. Gills; they have many thin filaments and lamellae with a large surface area, and use countercurrent flow of blood and water to maximise oxygen uptake.

3. Oxygen debt is the extra amount of oxygen the body needs after vigorous exercise to break down the lactic acid that accumulated during anaerobic respiration.

4. RQ = 20 ÷ 20 = 1.0; this value suggests the substrate being respired is a carbohydrate.

5. Any two: age, sex, level of physical activity, body size, climate.

End of Topic Review Question Answers

Topic 1: Cell Biology and Biodiversity

1. Biology is the study of living organisms; applications include agriculture and medicine (any two).

2. E.g. Botany–Botanist, Zoology–Zoologist, Genetics–Geneticist (any three correct pairs).

3. Should: interest, ability. Should not: gender, culture, disability, environment, stereotypes (any two each).

4. E.g. sweep net (catching flying insects), pitfall trap (trapping crawling insects), forceps (handling specimens), hand lens (observing specimens).

5. Press the plant → dry it → mount on a sheet → label (name, date, locality) → store/protect.

6. Light microscope uses light and has lower magnification/resolution; electron microscope uses electrons and has much higher magnification/resolution, requiring dead specimens.

7. Plant cell has a cell wall, chloroplasts and a large permanent vacuole; animal cell lacks these and has an irregular shape (any three).

8. A red blood cell is biconcave (increasing surface area for gas exchange), has no nucleus (more room for haemoglobin) and contains haemoglobin which binds oxygen for transport.

9. Cell → Tissue → Organ → Organ system → Organism.

10. Biuret test; the solution turns from blue to purple/violet in the presence of protein.

11. An enzyme is a biological catalyst (protein) that speeds up chemical reactions without being used up.

12. Temperature, pH, substrate concentration, enzyme concentration.

13. High temperature causes the enzyme's protein structure to change shape permanently (denature), so it can no longer bind its substrate and catalyse the reaction.

14. Water acts as a solvent and transport medium and helps regulate body temperature; mineral salts support specific processes e.g. calcium for bones, iron for haemoglobin.

15. It helps consumers check the quantity and safety of ingredients, preservatives, colourings and the expiry date before buying, enabling informed and healthy food choices.

Topic 2: Anatomy and Physiology of Plants

1. Autotrophic nutrition: organism makes its own food, e.g. a green plant through photosynthesis. Heterotrophic nutrition: organism depends on others for food, e.g. dodder (parasitic).

2. The chloroplast has a double membrane, fluid stroma, and grana (stacks of thylakoids) containing chlorophyll; the grana trap light energy (light stage) while the stroma is where carbon dioxide is converted to glucose (dark stage).

3. Photosynthesis produces food and oxygen, removes carbon dioxide from the air, and forms the base of food chains.

4. In monocots, vascular bundles are scattered; in dicots, they are arranged in a ring.

5. Root pressure pushes water up from the roots; capillarity allows water to rise in narrow xylem vessels; transpiration pull draws water up as it evaporates from the leaves (cohesion-tension theory).

6. Transpiration is the loss of water vapour from aerial parts of a plant, mainly through stomata. Factors: light intensity, temperature, humidity, wind speed (any four).

7. Translocation is the movement of manufactured food (sucrose) from the leaves (source) through the phloem to other parts of the plant (sink) such as roots and fruits.

8. Stomata (leaf surface), lenticels (woody stems), cuticle (leaf surface, minor), pneumatophores (mangrove aerial roots).

9. Guard cells become turgid (open pore) when they absorb water, and flaccid (close pore) when they lose water, controlled by movement of potassium ions.

10. Aerobic respiration requires oxygen and releases more energy, producing CO2 and water; anaerobic respiration occurs without oxygen, releases less energy, and produces ethanol and CO2 in plants.

11. Baking (bread rising), brewing (alcohol production), biogas production, silage making (any three).

12. On hot, windy afternoons, higher temperature and wind speed increase the rate of transpiration (water loss) beyond what the roots can replace, causing the plant to wilt; cool, calm mornings have lower transpiration rates.

Topic 3: Anatomy and Physiology of Animals

1. The mosquito has a needle-like piercing and sucking proboscis that penetrates the skin and draws blood/fluids, matching its feeding habit.

2. The eagle has a strong, hooked beak adapted for tearing flesh, suited to its diet as a predator.

3. Any three: carries digested food/oxygen/hormones to cells, removes waste, distributes heat, transports defence cells.

4. Open system: blood flows freely in the body cavity, e.g. insects. Closed system: blood remains in vessels, e.g. mammals/fish.

5. Deoxygenated blood enters the right atrium, passes to the right ventricle, and is pumped to the lungs; oxygenated blood returns to the left atrium, passes to the left ventricle, and is pumped to the body. Valves prevent backflow.

6. The lymphatic system collects excess tissue fluid and returns it to the blood, and its lymph nodes contain lymphocytes that help defend the body against disease.

7. Platelets gather at the wound site and trigger a reaction converting fibrinogen to fibrin, which traps blood cells to form a clot.

8. Blood group O can safely be given because it has no A or B antigens on its red cells (universal donor), so it is unlikely to cause a harmful reaction in a patient with blood group A.

9. Large surface area, thin walls, moist surface, good blood supply.

10. Oxygen diffuses from the air in the alveolus into the blood in the surrounding capillaries, while carbon dioxide diffuses from the blood into the alveolus to be exhaled; this occurs because of the concentration gradients across the thin, moist alveolar walls.

11. Aerobic respiration uses oxygen and releases more energy (CO_2 + water produced); anaerobic occurs without oxygen, releases less energy, and produces lactic acid in animals.

12. Oxygen debt is the extra oxygen needed after exercise to break down lactic acid built up during anaerobic respiration; e.g. a sprinter breathes heavily after a race to repay this debt.

13. RQ = volume of CO_2 released ÷ volume of O_2 consumed. Approximate values: carbohydrates ≈ 1.0, fats ≈ 0.7, proteins ≈ 0.8.

14. Any four: age, sex, level of physical activity, body size, climate.